Ship Real AI

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The AI Engineer Bootcamp for LLMs, RAG, MCP,
AI Agents and Evals, from First Principles
to Production and Interviews

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Draft

Ship Real AI: The AI Engineer Bootcamp for LLMs, RAG, MCP, AI Agents and Evals, from First Principles to Production and Interviews

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Preface

The preface is written last, once the modules are final.

PART I

The job and the foundations

These modules explain what a model can and cannot do, and everything an AI engineer builds around it.

01

What an AI engineer builds

A language model can write a polite reply in Hindi, explain a return policy, or sort a thousand complaints in a few minutes. Yet when a company puts one in front of real customers, most of the work has little to do with the model itself. This module explains why. It starts with what a model can and cannot do, and ends with a map of everything an AI engineer builds around it. That map is the plan for the rest of the book.

1.1 What does a language model actually do?

You send it text, and it sends back text. That is all a single call to a model does.

GPT, Gemini, Claude, Llama, Qwen and DeepSeek are names of models. It helps to keep three words apart. The **model** is the trained program that turns text into text. The **model provider** is the company that runs it and sells access, such as OpenAI, Google or Anthropic. The **application** is a product built on top, such as the ChatGPT app. ChatGPT is not a model; it is an application that uses OpenAI's models and adds a lot around them.

Now ask a few simple questions about what a model can do on its own.

Ask it about an order placed this morning. It cannot answer. A model learns from data collected up to a certain date, and knows nothing after that. It has also never seen your company's database. This limit is called the **knowledge cutoff**.

Ask a question, then a follow-up such as "and how long will that take?" It does not know what "that" means. Every call to a model is separate, and the model keeps nothing from one call to the next. Chat apps seem to remember because the app sends the earlier messages again with every new one. Module 4 builds this.

Paste in three hundred pages of policy documents and a year of emails. It will refuse, or fail. Each call accepts only a limited amount of text, called the **context window**. Module 2 explains what fills it and why a longer input does not always give a better answer.

Ask the same question twice. You can get two different answers. An ordinary program gives the same output for the same input, every time. A model picks its words by prob-

ability, so its output can change from one call to the next (Module 2 shows how). This is the biggest difference between AI systems and ordinary software, and much of this book deals with its effects.

Ask it to check an order, send an email or issue a refund. It cannot. It can only write text. It can write "I have issued your refund", but writing that sentence does not move any money.

1.2 Three kinds of AI work

People who "work in AI" usually do one of three jobs. Some **make models**: they collect data and train models, which needs large research teams and large budgets. Some **serve models**: they make trained models run fast and cheaply on hardware, which is called inference engineering. And some **build applications**: they take a model that already exists and turn it into a product that works for real users.

This book is about the third job. You will learn enough about how models work to use them well, but you will not train a large model.

1.3 If the model cannot act, what can?

In software, what does the actual work, such as looking up an order or sending an email? Code. A function such as `check_order_status(order_id)` can query a database and return the answer.

Can that function understand a customer who writes "mera parcel kab aayega"? No. It needs an exact order number, passed in exactly the right way. The model understands the customer perfectly, but cannot query anything. Each one has exactly what the other lacks.

So put them together. The model reads the customer's message and decides which function should run and with which values. Your program runs the function. The result goes back to the model, which writes the reply (Figure 1.1). A model connected to functions it can ask your program to run is called an **agent**, and each such function is called a **tool**.

Now look at one particular kind of tool: one that fetches information, such as a search over the company's policy documents. The model asks for the relevant paragraph, the paragraph is added to the question, and the model answers using it. That is **retrieval-augmented generation**, or **RAG**. The name describes the three steps in order: *retrieve* the information, *augment* (add to) the question with it, then *generate* the answer. RAG

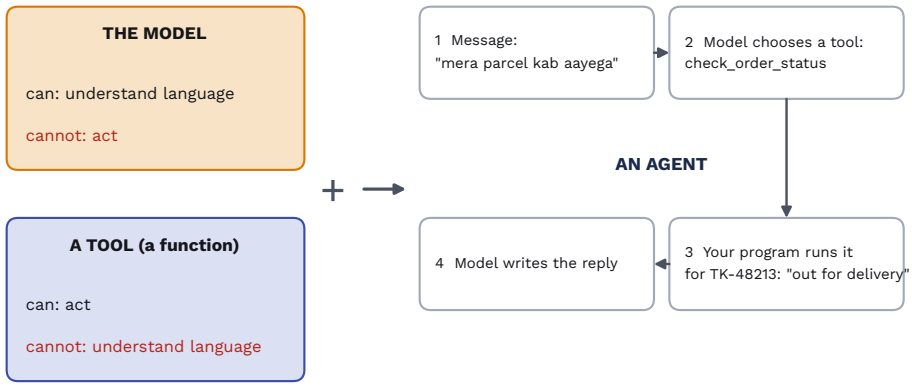


Figure 1.1 A model understands language but cannot act. A tool can act but cannot understand language. Connected, the model chooses the tool and your program runs it.

is not a separate technology. It is an agent whose tool fetches facts. Modules 6 to 8 are about doing it well.

You will also see agents described in other ways. “Model plus tools” came first. Then people noticed that what you put in front of the model decides most of the result, and started saying “model plus context”. The newest description is “model plus harness”, where the **harness** means everything around the model: the instructions, the tools, the memory, the checks, the code that runs the loop. All three descriptions are true. Building that harness is the job.

1.4 What does an AI engineer build around the model?

Figure 1.2 shows the parts, in the order a customer’s message passes through them. Each part has a module of its own.

1. **The front door** receives the message, and limits how many requests come in (Module 16).
2. **Safety checks** look at what comes in: private data, abuse, hidden instructions (Module 12).
3. **Control** decides what happens next. A fixed sequence of steps, or an agent that chooses, or a plan for a long task (Modules 11 and 14).
4. **The model** is called, perhaps several times (Modules 2 and 10).
5. **Context** decides what the model sees on each call: the instructions, the retrieved documents, the memory of the conversation (Modules 3, 4 and 6 to 8).
6. **Tools** let it act, through functions you write or through standard connectors such as MCP (Modules 5 and 13).

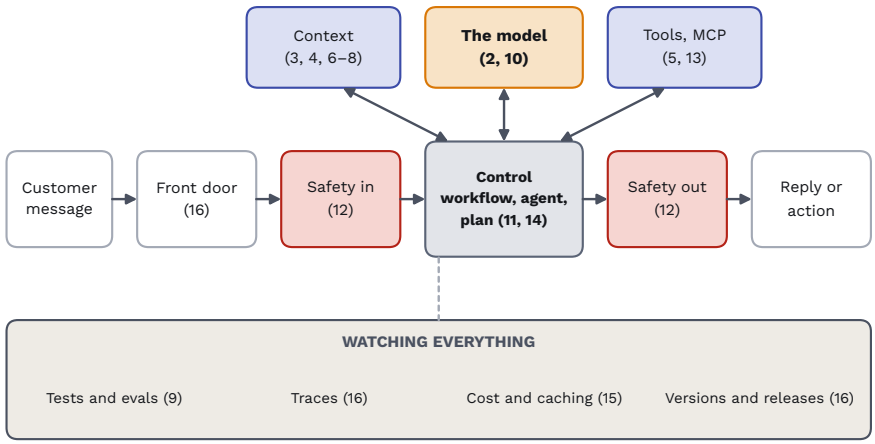


Figure 1.2 The parts of an AI system. A message enters at the front door, is checked, is handled by the control layer with the model, its context and its tools, and is checked again on the way out. Tests, traces, cost and caching watch over everything. The numbers are the modules that cover each part.

7. **Safety checks** look at what goes out, before the customer sees it or a refund is issued (Module 12).
8. Under all of it, **quality and operations**: tests that show whether it works, traces of every run, cost and caching, safe releases (Modules 9, 15 and 16).

The model is one box out of eight. The other seven are what the engineer builds.

1.5 Calling a model is not calling a function

In code, a model call looks like any other function call: text goes in, text comes out. It behaves very differently.

	An ordinary function	A model call
Speed	Microseconds	Seconds
Failure	Rarely fails	Fails sometimes: busy servers, limits, timeouts
Same input	Same output, every time	A different output is possible every time
Cost	Free to run	Charged for every call, by the amount of text

Each row changes how you write code around the model. Because the output varies, you check every reply before your code uses it (Module 5), and you cannot test it the usual way (Module 9). Because calls fail, you retry some failures and not others (Module 11). Because calls are slow, you run many at once, within the provider's limits (Module 15). And because every call costs money, you count what you send (Modules 3 and 15).

1.6 Should this problem use AI at all?

The table above gives a simple rule. A model is slower, costlier and less predictable than ordinary code. So when ordinary code can do the job, use ordinary code.

"Was this order placed more than 7 days ago?" is a date comparison. One line of code answers it correctly every time, for free. "Is this customer angry, and are they asking for a refund?", written in a mix of Hindi and English with spelling mistakes, cannot be written as a rule. That is where a model earns its cost.

Before building, ask four questions:

1. Is the input messy language, or a judgement that no rule can capture? If not, write the rule.
2. If the model is wrong, will someone notice, and how much will it cost?
3. What does success look like, as a number you can measure before and after?
4. Can you buy this ready-made? A generic support bot may be enough. You build your own when you need your own data, your own rules, or control over what it is allowed to do.

1.7 Your first call, and what it costs

Here is a complete call to a model in Python, using Anthropic's client library. Other providers' libraries look very similar.

```python
import anthropic

MODEL = "claude-opus-5-5"          # model names change; keep this in one place
client = anthropic.Anthropic()     # reads the API key from the environment

reply = client.messages.create(
    model=MODEL,
    max_tokens=1000,
    messages=[{"role": "user",
               "content": "Can I return opened earbuds after 10 days?"}],
)
for block in reply.content:
    if block.type == "text":
        print(block.text)
print(reply.usage.input_tokens, reply.usage.output_tokens)
```

The answer will be polite and sensible, and probably wrong for your shop, because the model has never seen your return policy. The last line prints how much text went in and came out, counted in **tokens**, the small pieces of text that models read and charge by. Module 2 explains what a token is.

Now work out the cost. Suppose the provider charges ₹350 for every million input tokens and ₹1,700 for every million output tokens. (These are assumed example prices. Output usually costs several times more than input, and Module 2 explains why.)

1. This question was about 30 tokens in and 120 tokens out: 30 × 350 ÷ 10,00,000 + 120 × 1,700 ÷ 10,00,000, about ₹0.21.
2. A real support request also carries instructions and the relevant part of the policy: say 3,000 tokens in and 300 out. That is 1.05 + 0.51, about ₹1.56 per message.
3. At 4,000 messages a day, that is about ₹6,240 a day, or about ₹1,87,000 a month.

Two things stand out. Most of the cost comes from the text sent *with* each question, not from the question itself. And the bill grows with every message, so a small saving per call becomes a large saving per month. Modules 3 and 15 come back to both.

1.8 The project for this book

Through the book, you will build one system: a customer-support agent for **Tarkika**, an imagined online marketplace that sells phones, electronics, fashion and home products across India. By the end, it will:

- answer questions from the company's policy documents, with the source of each answer, and say "I don't know" when the documents do not cover a question;
- look up orders, and prepare refunds that a person approves above a set amount;
- remember a customer's earlier messages, and keep each customer's data separate;
- refuse hidden instructions in customer emails;
- report its own quality, cost and speed.

Each of these is one part of Figure 1.2. The first thing to understand is the model in the middle: how it reads text, why the same question can get different answers, and why you pay for every piece of text you send. That is Module 2.

02

How LLMs work, for engineers

You do not need to know how to build a model to use one well. But a few things about how a model works explain most of what you will see in practice. This module shows them, mostly in pictures. Where a free online tool lets you try something yourself, the module points you to it.

2.1 Tokens: what the model actually reads

A model does not read words the way we do. It reads small pieces of text called **tokens**. Figure 2.1 shows how a real tokenizer, the one used by OpenAI's GPT-4o models, splits a few words.

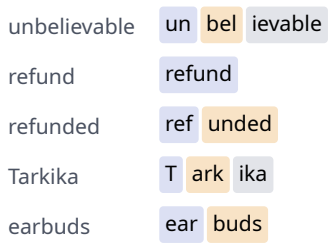


Figure 2.1 Real splits from the GPT-4o tokenizer. Common words are one token; longer or rarer words are cut into pieces.

"Refund" is common, so it is one token. "Refunded" is two. "Unbelievable" is three. A made-up name like "Tarkika" is cut into "T", "ark" and "ika", because the model has never seen it as a whole. In English, 100 tokens is roughly 75 words.

Why should you care? Because everything about a model is counted in tokens. The price is per token. The speed is per token. The limit on how much you can send is in tokens.

Do not confuse a token with a **chunk**. A token is a tiny piece the model reads. A chunk is a large piece of a document, often a few hundred tokens, that you cut out yourself for searching. Chunks come in Module 7.

Try it: search for “OpenAI tokenizer” or open tiktokenizer.vercel.app. Paste any sentence and watch it split into coloured pieces, exactly like the figures here.

2.2 The same message in five languages

Now look at Figure 2.2. It is one message, “My refund has not come yet”, written five ways, and split by the same tokenizer. A dot marks a space.



Figure 2.2 One message, five ways, split by the GPT-4o tokenizer. Each coloured box is one token. The dotted circles in Hindi, Telugu and Kannada are pieces of a letter that the tokenizer cut away from the rest.

English needs 6 tokens. For the same meaning, Telugu needs 13, Hindi 14 and Kannada 16. In these scripts, the tokenizer often cuts a single letter into two pieces, like the vowel signs standing alone in their own boxes.

Why? The tokenizer learned its list of pieces mostly from English text. Common English words got their own tokens. Words in Indian scripts were seen less often, so they are built from many small pieces.

Figure 2.3 shows that this is improving, but not gone. The older GPT-4 tokenizer needed 29 tokens for the Hindi message, 36 for Telugu and 46 for Kannada. Studies across many languages find gaps of three to five times for Hindi, and up to fifteen times between some languages (*Reducing Tokenization Premiums for Low-Resource Languages* 2026; Petrov et al. 2023).

For your work, more tokens for the same message means a higher bill, a slower reply, and less room for anything else in the request. Hinglish, written in English letters, sits in between. If your users write in Indian languages, count a sample of their real messages before you plan a budget.

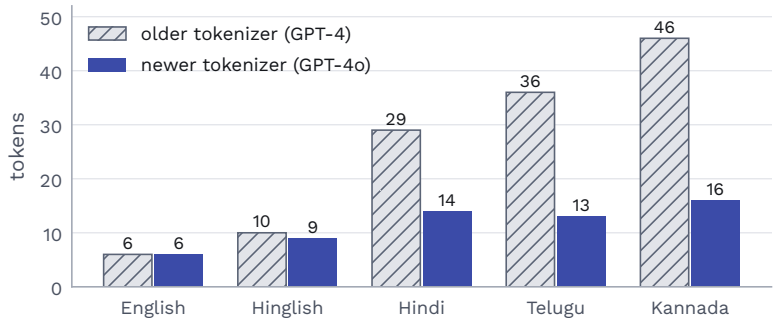


Figure 2.3 Tokens for the same message, counted with the older GPT-4 tokenizer and the newer GPT-4o tokenizer.

2.3 Watching the model write

A model writes one token at a time. At each step it looks at all the text so far, gives every possible next token a chance, picks one, and adds it. Then it repeats. Figure 2.4 shows four steps.

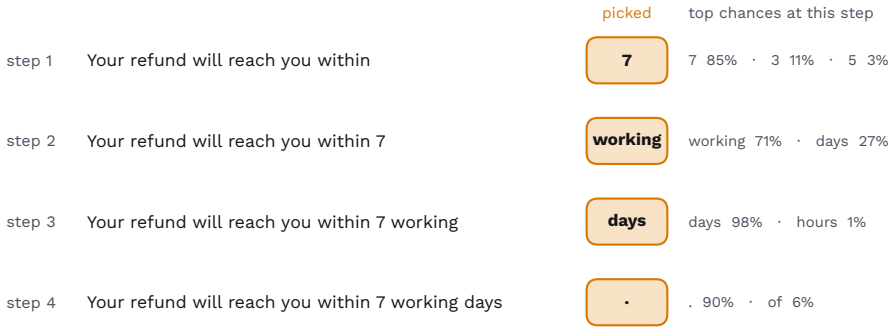


Figure 2.4 The model writing a reply, one token per step. At each step it picks one token from many possible ones. (Chances made up for illustration.)

At step 1, “7” had an 85% chance, so it was picked. But “3” had 11%. Run the same request again, and sometimes the reply will say 3 days, not 7. Nothing is broken. The model is picking by chance, and the chances can go either way.

A setting called **temperature** controls how strongly the model prefers its top choice (Figure 2.5). At a low temperature, the top token wins almost every time. At a high temperature, the chances spread out and replies vary more.

This is why the same question can get a different reply each time, and why your code must never expect one exact reply.

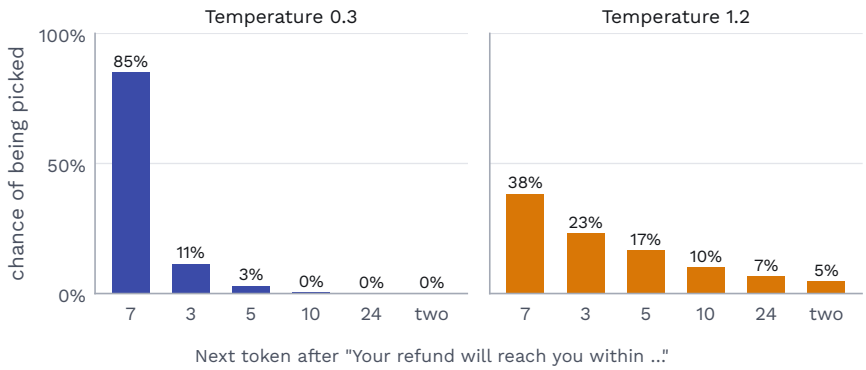


Figure 2.5 The same chances at two temperatures. Low temperature makes the top choice dominate; high temperature spreads the chances out. (Numbers made up for illustration.)

2.4 How a model learns

A model starts knowing nothing. Training teaches it with one simple exercise: show it some text with the next word hidden, and ask it to guess. If the guess is wrong, its internal numbers are adjusted a little, so the next guess would be better. This is repeated with trillions of words. Slowly, those numbers come to hold grammar, facts and patterns.

That is all training is: guess, check, adjust, repeat.

The internal numbers are called **parameters**, and a large model has billions of them. For your work, this matters in one way: memory. Each parameter usually takes 2 bytes.

1. A model with 8 billion parameters needs about 16 GB of memory just to load.
2. Shrink each parameter to half a byte, and it needs about 4 GB.
3. Training the same model needs roughly 8 times more memory than running it, about 128 GB (Rajbhandari et al. 2020).

This is why most companies use models that someone else has trained. Running a model is within reach. Training a large one is not.

2.5 How a word gets its meaning from its neighbours

“Bank” means the side of a river in “the river bank”, and a place for money in “my bank account”. The model works out which meaning applies the same way you do: by looking at the words around it. In Figure 2.6, “bank” pays most of its attention to “river”.

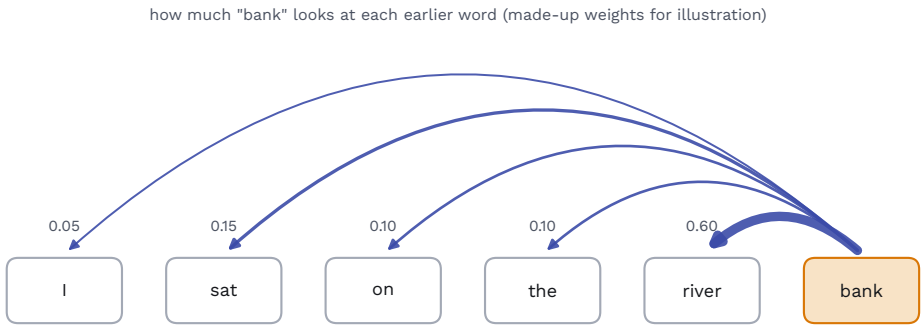


Figure 2.6 Each word looks at the other words to understand itself. Here "bank" looks mostly at "river". (Numbers made up for illustration.)

This idea is called **attention**. It is the heart of the **transformer**, the design almost every modern model uses.

Attention has a cost you will meet in practice. To write each new word, the model looks back at every word before it. To save time, it keeps notes on the earlier words in memory, and the notes grow with every word. For a typical 8-billion-parameter model:

1. One conversation of 8,000 tokens needs about 1 GB of notes.
2. Fifty such conversations at once need about 50 GB, three times the memory of the model itself.

That is why long conversations are expensive, and why every model limits how much text it accepts.

Try it: Transformer Explainer (poloclub.github.io/transformer-explainer) runs a small model in your browser. Type a sentence and watch the attention lines move between words, then see the chances for the next token (Cho et al. 2025).

2.6 From reading the internet to being helpful

A model is trained in stages (Figure 2.7).

1. **Pretraining.** The model guesses the next word across a huge amount of text from the web, books and code. Meta reports about 15 trillion tokens for its Llama 3 models (Llama Team, AI at Meta 2024). The result knows a lot, but only continues text. Ask it "What is your return policy?" and it may reply with more questions, because that is what often follows a question on the web.
2. **Fine-tuning.** It sees many examples of an instruction followed by a good answer. Now it answers instead of just continuing.

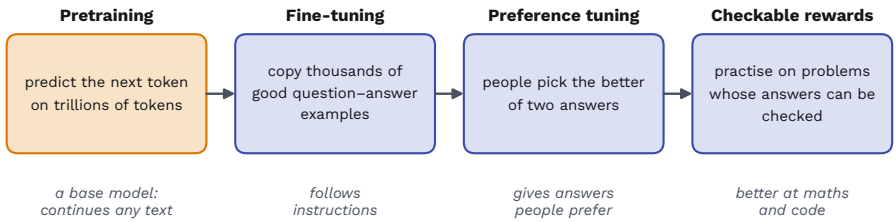


Figure 2.7 The stages of training. Each stage starts from the model the previous stage produced.

3. **Preference tuning.** People compare two answers and pick the better one. The model learns to give answers people prefer (Ouyang et al. 2022).
4. **Checkable rewards.** It practises on problems whose answers can be checked automatically, such as maths, and code with tests (DeepSeek-AI 2025). This is why recent models are so strong at maths and code.

Step 3 has a side effect. People tend to prefer answers that agree with them and sound sure. So models learn to agree and to sound confident, even when they are wrong (Sharma et al. 2023). When a model says "I am sure", treat it as a tone of voice, not a fact. Module 10 shows how to get confidence numbers you can trust.

2.7 Meaning as a map

"I want my money back" and "refund chahiye" share no words, yet mean the same thing. A search for matching words would never link them.

An **embedding** turns a piece of text into a long list of numbers, so that texts with similar meaning get similar lists. Think of each list as a point on a map. In Figure 2.8, refund messages land in one area, delivery complaints in another, and product questions in a third, whatever language they are written in.

To check whether two messages mean the same thing, you measure how close their points are. The usual measure is **cosine similarity**, which gives a score near 1 for very similar meanings.

How long should the list of numbers be? Longer lists capture finer differences in meaning, but take more space and more time to search. A million stored pieces of text take about 6 GB with 1,536 numbers each, and about 1.5 GB with 384. Pick the smallest size that still finds the right results on your own data.

Embeddings come from separate, smaller models that only produce these lists. Module 6 builds search on top of them.

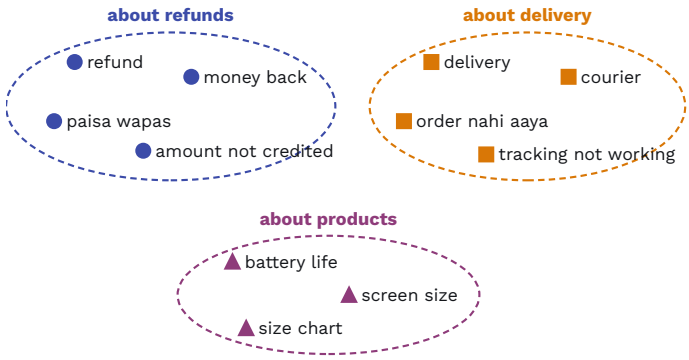


Figure 2.8 Messages placed by meaning. Real embeddings have hundreds of numbers; this picture squeezes them into two so they can be drawn. (Positions illustrative.)

Try it: the Embedding Projector (projector.tensorflow.org) shows thousands of words as points in 3D. Search for a word and see which words sit closest to it.

2.8 More text can mean worse answers

Every model has a **context window**: the most tokens it can take in one call. Some now hold a million tokens or more. So it is tempting to send everything: all the policies, all the past emails, every order.

Look at Figure 2.9. When a 30-token question sits beside 300 tokens of material, it is 9% of what the model reads. Beside 300,000 tokens, it is 0.01%. The question is still there, but it is a tiny part of what the model's attention is spread over.

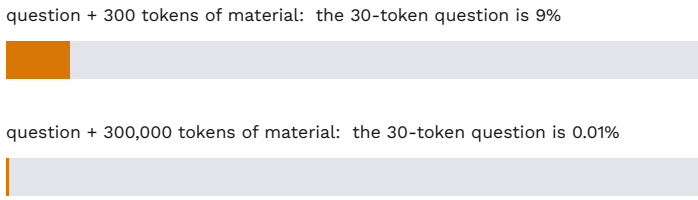


Figure 2.9 How much of the input is the actual question, with a little material and with a lot.

Experiments show the result. As the input grows, answers get worse, even on simple tasks, and even far below the window's limit (Hong et al. 2025). This is called **context rot**.

So a big window does not mean you can stop choosing what to send. Module 3 is about choosing well.

2.9 Reading is fast, writing is slow

Figure 2.10 shows the difference between how a model handles your input and its own output.

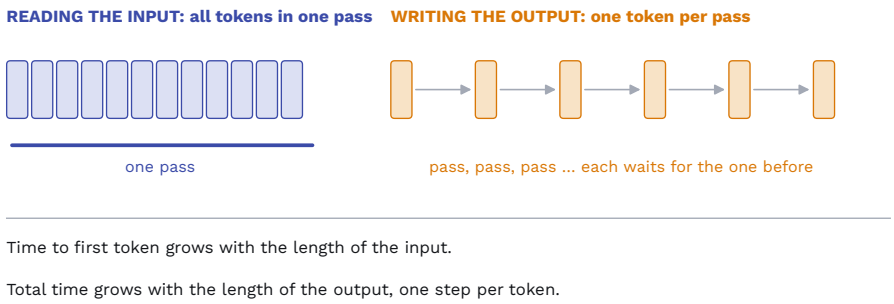


Figure 2.10 The model reads all the input in one go, but writes the output one token at a time.

Your input is known from the start, so the model reads all of it in one go. Its output is not. It must write token 1 before it can write token 2, so it works through the output one token at a time. Each output token takes much more computer time than each input token. That is why providers charge around four to five times more for output.

You will notice two speeds:

- **Time to first token**: how long before anything appears. It grows with the length of your input, because all of it is read first.
- **Tokens per second**: how fast the rest arrives once it starts.

Showing the reply word by word as it is written, called **streaming**, makes a slow reply feel fast, because the user starts reading early.

Providers use a few tricks to make all this cheaper. They process many users' requests together (**batching**). They reuse the work when many requests start with the same text (**prefix caching**; Module 3 shows how to benefit). And they store the model's numbers in fewer bits to save memory and time (**quantisation**), at a small cost in quality.

2.10 Same question, different answer, even at temperature 0

At temperature 0, the model should always pick its top token, so the same question should always get the same answer. Often it does not.

The reason is in how providers run models. Your request is processed together with other people's requests, and the size of that group changes every moment. That

changes the order in which the computer adds up its numbers, which can change the result in the last few digits. Usually this changes nothing. But when two tokens are almost tied, the tiny difference can flip the choice, and the rest of the reply goes another way (He and Thinking Machines Lab 2025).

The lesson is simple: never build anything that needs the model to give exactly the same text twice. Check the meaning of the answer instead. Module 9 shows how.

2.11 Thinking costs tokens

What is 2 + 2? You answer at once. What is 17 × 24? You probably work it out in steps. Models are similar: on problems with several steps, they do better if they write out their working first.

Thinking models do this automatically. Before the answer, they write hidden reasoning, and a setting usually called **effort** controls how much. You pay for that reasoning like any other output. Figure 2.11 shows what that means for a simple job, sorting a message into one of four categories.

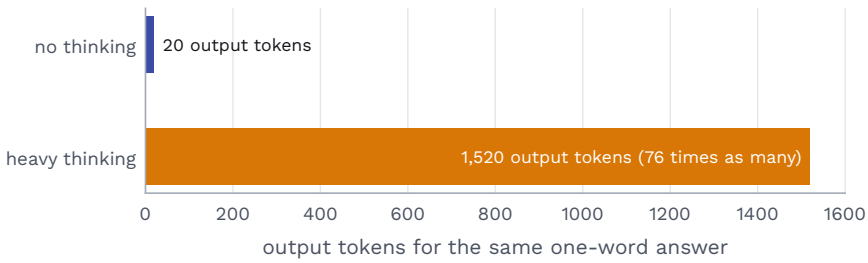


Figure 2.11 Output tokens for sorting one message, with and without heavy thinking. The answer is the same one word.

The answer is the same one word, at 76 times the output cost. For sorting messages, that is waste. For working out a refund across three partial returns and a coupon, the thinking may be what makes the answer right. So compare models by the **cost per finished task**, not by the price per token.

2.12 Big models that still run fast

A bigger model usually gives better answers. But every word passes through the whole model, so a bigger model is also slower and costs more per word.

A **mixture of experts** gets around this. The model is split into many parts, called experts, and for each token a small router sends the work to only a few of them (Figure 2.12). Mixtral 8x7B, for example, has about 47 billion parameters in total, but uses about 13 billion for each token (Jiang et al. 2024).

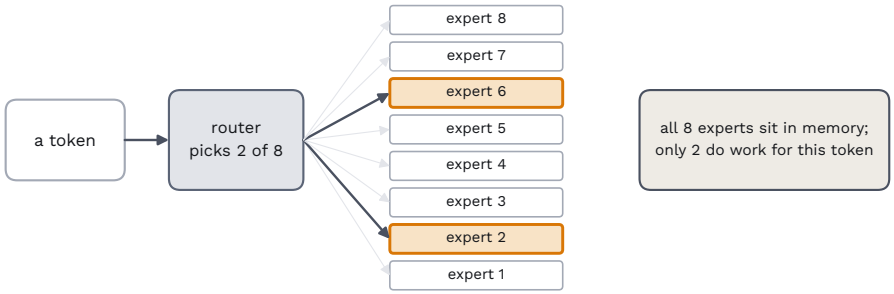


Figure 2.12 A mixture of experts. Each token goes to only two of the eight experts, but all eight must be kept in memory.

The catch: speed depends on the part that is used, but memory depends on the whole. Every expert must be loaded, because the next token might need a different one. Such a model can be as fast as a small model and still need the memory of a large one.

2.13 Images, documents and speech

The same design handles more than text. An image is cut into small squares, and each square is treated like a token, so the model can answer a question about a photo. A scanned page can be read the same way. Speech is cut into short slices of sound, or turned into text first by a separate model.

Images are not free. One image can cost as many tokens as a full page of text, so a screenshot with every question adds up quickly.

2.14 Choosing a model

There is no single best model. There is only the right model for each job.

- **Sorting, extracting and short answers**: a small, fast, cheap model. A large one gives the same answer at many times the cost.
- **Problems with several steps, or writing that must be exact**: a large model, sometimes with thinking turned on.
- **Open-weight models**, which you can download and run yourself: data stays in-house, and they can be cheaper at high volume. But you run the hardware, and "open" does not always mean free for commercial use. Read the licence.
- **Models through an API**: no hardware to manage. Usually the best place to start.

The only reliable way to choose is to test. Try a few models on your own real examples, and compare quality, speed and cost per finished task. Module 9 shows how.

The model knows only what is in front of it at the moment you call it. So what should you put in front of it, and in what order? That is Module 3.

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